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Question 5 => Pseudocode

Start
totalBooks = 4
bookIDs[4] = {101, 102, 103, 104}
bookTitles[4] = {"Programming in C", "Data Structures", Database systems", "Computer Networks"}
availablecopies[4] = {5, 3, 0, 4}

name = ""
studentID = 0
numberType = 0 (1 for Student, 2 for faculty)
borrowedBookIDs[3] = {0, 0, 0}
borrowedCount = 0
totalfine = 0.0
fineafterwaiver = 0.0

CALL registermember(name, studentID, membertype)
Do
  Call displayMenu()
  Print "Enter choice:"
  Read choice

  Switch choice
    Case 1:
      Call registermember(name, StudentID, memberType)
      Break
    Case 2:
      Call displayBooks(bookIDs, bookTitles, availablecopies, totalBooks)
      Break
    Case 3:
      Callborrowbook(bookIDs, bookTitles, availablecopies, totalbooks, borrowedbooksIDs, borrowedcount, membertype)
      Break

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Case 4:
Call returnBook (bookIDs, availableCopies, totalBooks, borrowedBookIDs, borrowedCount, memberType, totalFine)
fineAfterWaiver = totalFine
Break
Case 5:
Call displayMemberRecord (name, studentID, memberType, borrowedBookIDs, borrowedCount, totalFine, fineAfterWaiver)
Break
Case 6:
Print "Exiting program. Thank you!"
Break
Default
Print: "Invalid choice! Please try again."
END Switch
While choice != 6
END

// Function to display menu options
Function displayMenu()
Print "== University Library Management System =="
Print "1. Number Registration"
Print "2. Display Available Books"
Print "3- Borrow a Book"
Print "4. Return a Book"
Print "5. Display Member Record"
Print "6. Exit"
END function

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```
// Function to register a member
Function registerMember(name, studentID, membertype)
    Print "Enter member name:"
    Read name
    Print "Enter Student / Member ID:"
    Read student ID
    member type = 0
    While memberType != 1 AND membertype != 2
        Print "Entermembership type (1 for Student, 2 for faculty)."
        Read member type
        If membertype != 1 AND membertype != 2 then
            Print "Invalid membership type! Enter 1 or 2."
        End If
    End While
    Print "Member registered successfully!"
END function

// Function to Display available books
Function displayBooks (bookIDs, bookTitles, availableCopies, totalBooks)
    Print "--------"
    Print "ID \t\t Title \t\t Copies"
    Print "--------"
    For i=0 to totalBooks - 1
        Print bookIDs[i] + "\t" + bookTitles[i] + "\t" +
                                    availableCopies[i]
    END For
    Print "----------"
END Function
```


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```
// Function to borrow a book
Function borrowBook(bookIDs, bookTitles, availableCopies, totalBooks,
    borrowedBookIDs, borrowedCount, memberType)
If borrowedCount >= 3 then
    Print "Error: You have reached the max limit of 3
        borrowed books!"
    Return
End If
Print "Enter book ID to borrow:"
Read b_id
index = -1
For i = 0 to totalBooks - 1
    If bookIDs[i] == b_id then
        index = i
        Break
    End if
End For
If index == -1 then
    Print "Error: Invalid Book ID!"
Else If availableCopies[index] <= 0 then
    Print "Error: Book is currently out of stock!"
Else
    alreadyHas = 0
    For j = 0 to borrowedCount - 1
        If borrowedBookIDs[j] == b_id then
            alreadyHas = 1
        End If
    End For
    If alreadyHas == 1 then
        Print "Error: You have already borrowed this book!"
    Else
        borrowedBookIDs[borrowedCount] = b_id
```


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borrowed Count + = 1
available Copies [index] - - = 1
Print " Books successfully borrowed!"
END If
END If
END Function

// Function to return a book and handle fines
Function returnBook (bookIDs, available Copies, total Books, borrowed bookIDs, borrowed Count, member Type, total Fine)
If borrowed Count == 0 then
Print "Error: You have no books to return!"
Return 0.0
End If
Print "Enter book ID to return!"
Read b-id
pos = -1
For i = 0 to borrowed Count - 1
If borrowed Book IDs [i] = - b-id then
pos = i
Break
End-If
End For
If pos == -1 then
Print " Error: You have not borrowed a book with this ID!"
Return 0.0
End If
Print " Enter Actual number of days the book was kept "
Read days-kept

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```
While days_kept <= 0
    Print "Invalid days! Enter a positive number"
    Read days_kept
End While
allowed_days = 14
If member_type == 2 then
    allowed_days = 30
End If
fine = 0.0
If days_kept > allowed_days then
    extra_days = days_kept - allowed_days
    If member_type == 1 then
        fine = extra_days * 20.0
    Else
        fine = extra_days * 10.0
    End If
End If
Print "Fine calculated: Rs." + fine
total_fine += fine
Print "Enter membership code for waiver"
Read code_input
waiver_applied = 0
If code_input == "LIB2026" then
    fine = fine * 0.5
    Print "50% waiver applied!"
    waiver_applied = 1
End If
inv_index = -1
For i = 0 to total_books - 1
    If bookIDs[i] == b_id then
        inv_index = i
        Break
    End If
End For
```

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```
If inv_index != -1 then
    available copies[inv_index] += 1
End If
For i = pos to borrowedCount - 2
    borrowedBookIDs[i] = borrowedBookIDs[i+1]
End For
borrowedCount -= 1
Print "Book returned successfully"
Return fine
END function
```

```
// Function to Display member record
Function displayMemberRecord(name, studentID, membertype,
borrowedBookIDs, borrowedCount, totalfine, fine
fineafterwaiver)
Print "=== Member Record ==="
Print "Name: " + name
Print "ID: " + studentID
If membertype == 1 then
    Print "Type: Student"
Else
    Print "Type: Faculty"
End If
Print "Currently Borrowed Book IDs:"
If borrowedCount == 0 then
    Print "None"
Else
    For i = 0 to borrowedCount - 1 Print BorrowedIDs[i]
    END for
END If
Print "Total fine charged before waiver: Rs." + totalfine
Print "Final fine payable after waiver: Rs." + fine after
END Function.                                          waiver
```


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